OMMC '26

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Contents

Problem 1 2
Problem 2 3
Problem 3 4
Problem 4 6
Problem 5 7
Problem 6 9
Problem 7 11
Problem 8 13
Problem 9 15
Problem 10 17

Problem 1. Let acute-angled $\triangle ABC$ have orthocenter H, nine-point center N and circumcenter O. If r is the inradius of $\triangle HBC$, prove that $OA + 2NA > 2HA + 4r$.

Solution. We proved the halved inequality. Let D be the foot from A to BC.

> **Claim** — $HA + 2r < AD$.

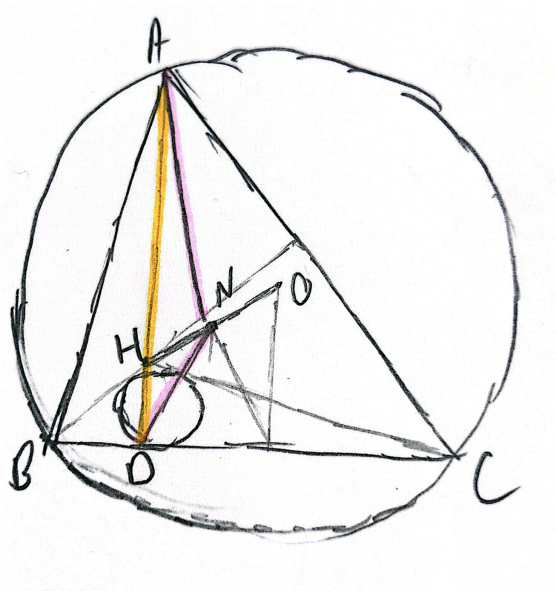
Proof. Note that $2r$ will be less than the height from H to BC which completes the proof. □

> **Claim** — $AD \leq NA + \frac{OA}{2}$.

Proof. Recall the well-known fact that the circumradius is two times the length of the nine point circle radii, which is because of a $\frac{1}{2}$-homothety at H. Thus, $\frac{1}{2}OA = ND$.

However, by Triangle Inequality, $AD \leq AN + ND$. This completes the proof. □

Comparing the two, we get our desired answer. □



Problem 2. Let m be a positive integer. For n a positive integer, let $f(m,n)$ be the number of ways to tile an $m \times 4n$ rectangular grid with tetrominoes, where reflections and rotations are considered distinct. Prove that $\lim_{n\to\infty} f(m,n)^{1/n}$ exists.

Solution. Let $a_n = f(m,n)$.

Claim — We have $a_{x+y} \geq a_x \cdot a_y$.

Proof. Consider the set of solutions in which the tetrominoes tile the first x columns and seperate tetrominoes tile the last y columns. This injectively maps to a subset of the set of tilings for $x+y$. This implies $a_{x+y} \geq a_x \cdot a_y$. Since $a_1 \geq 1$, this also implies a_x is increasing. Additionally, this implies

$$a_{\sum_{i=1}^{n} x_i} \geq \prod_{i=1}^{n} a_{x_i}.$$

□

Claim — We have $h = \limsup_{n\to\infty} a_n^{\frac{1}{n}}$ and $\ell = \liminf_{n\to\infty} a_n^{\frac{1}{n}}$ are finite real numbers.

Proof. This is equivalent to showing $a_n^{\frac{1}{n}}$ is bounded. Since $a_n \geq 0$, $a_n^{\frac{1}{n}} \geq 0$. Note that if we know which of each cell's neighbors are in the same tetromino as it, we know the tiling. Each cell has at most 4 possible neighbors, so it has at most 2^4 options for which of it's neighbors are in the same tile as it. Thus $a_n \leq 16^{4mn}$, so $0 \leq a_n^{\frac{1}{n}} \leq 16^{4m}$ for all n. □

Claim — We have $\ell = h$.

Proof. Assume otherwise, choose $M = \frac{\ell+h}{2}$, so $\ell < M < h$. We note that $M \geq 1$ because $a_1 \geq 1$. We know there must exist some i with $a_i^{\frac{1}{i}} \geq M \implies a_i \geq M^i$, and we know $a_{ki+r} \geq a_i^k \cdot a_r$, so for all n, we can write $n = ki + r$ with $1 \leq r \leq i$, so $a_n \geq a_i^k \cdot a_r$ so

$$a_n^{\frac{1}{n}} \geq a_i^{\frac{k}{n}} \cdot a_r^{\frac{1}{n}} \geq M^{\frac{ki}{n}} \geq M^{\frac{ki}{ki+i}} = M^{\frac{k}{k+1}}.$$

This gives us

$$\ell = \liminf_{n\to\infty} a_n^{\frac{1}{n}} \geq \liminf_{k\to\infty} M^{\frac{k}{k+1}} = M,$$

contradiction. □

Since $\limsup_{n\to\infty} a_n^{\frac{1}{n}} = \liminf_{n\to\infty} a_n^{\frac{1}{n}}$, this implies $\lim_{n\to\infty} a_n^{\frac{1}{n}}$ exists. □

Problem 3. Find all functions $f : \mathbb{R}^+ \to \mathbb{R}^+$ such that for all positive real numbers x and y,

$$xyf(xy) + yf(f(x) + y) + f(1 + xy) = xf(x) + yf(y).$$

Solution. The answer is $f(x) \equiv \frac{1}{x}$. For verification, we observe

$$1 + \frac{y}{\frac{1}{x} + y} + \frac{1}{1 + xy} = 1 + 1.$$

We seperate this solution into two main claims.

Claim — The limit

$$\lim_{x\to\infty} xf(f(1) + x) = f(1)$$

holds.

Proof. The assertion $x = 1$ gives

$$yf(f(1) + y) + f(1 + y) = f(1) \implies yf(f(1) + y) = f(1) - f(1 + y) \le f(1).$$

Shifting $y \to y + 1 - f(1)$, supposing $y > f(1) - 1$, we get

$$(y + 1 - f(1))f(1 + y) \le f(1) \implies f(1 + y) \le \frac{f(1)}{y + 1 - f(1)}.$$

Going back to $x = 1$, we arrive at the inequality

$$yf(f(1) + y) = f(1) - f(1 + y) \ge f(1)\left(1 - \frac{1}{y + 1 - f(1)}\right).$$

Combining the two inequalities,

$$f(1)\left(1 - \frac{1}{y + 1 - f(1)}\right) \le yf(f(1) + y) \le f(1).$$

Taking squeeze theorem on $y \to \infty$:

$$\lim_{y\to\infty} [yf(f(1) + y)] = f(1)$$

completes the proof. □

Claim — The two limits

$$\lim_{x\to\infty} f(x) = 0 \text{ and } \lim_{x\to\infty} xf(x) = f(1)$$

holds.

Proof. For the first limit, the previous claim clearly implies $f(f(1) + y) \to 0$ as $y \to \infty$. For the second limit:

$$\lim_{x\to\infty} [xf(x)] = \lim_{y\to\infty} [(y+f(1))f(y+f(1))] = \lim_{y\to\infty} [yf(y+f(1))]+\lim_{y\to\infty} [f(1)f(y+f(1))] = f(1).$$

□

Now, we have the tools to finish. Rearranging and fixing x,

$$\begin{aligned} xf(x) &= \lim_{y\to\infty} [xyf(xy) + yf(f(x)+y) + f(1+xy) - yf(y)] \\ &= f(1) + f(1) + 0 - f(1) \\ &= f(1), \end{aligned}$$

so $f(x) = \frac{f(1)}{x}$ for all x. Plugging this into the original equation gives

$$f(1) + \frac{xyf(1)}{f(1)+xy} + \frac{f(1)}{1+xy} = 2f(1),$$

which is equivalent to $\frac{xyf(1)}{f(1)+xy} = f(1) - \frac{f(1)}{1+xy} = \frac{xyf(1)}{1+xy}$, which is true if and only if $f(1) = 1$ and thus $f(x) = \frac{1}{x}$ is the unique solution. □

Problem 4. Find all ordered pairs of prime numbers (p, q) for which for all $n \geq 2$, there exist integers $a + b = 2n - 2$ and $p^a + q^b$ is an nth power.

Solution. Let p be prime.

The answer is $(p, q) = (2, 2)$. Verification is trivial because pick $a = b = n - 1$ and we get $p^a + q^b = 2^n$.

Take $n = 2$, so $p + q$ or $1 + p^2$ is a square. Clearly, $1 + p^2$ can not be a square due to sizing, so $p + q$ is a square.

Take $n = 3$, so $1 + q^4, p + q^3, p^2 + q^2$, and its symmetric counterparts are cubes. We rule out everything but $p^2 + q^2$:

- If $1 + q^4 = x^3$, this is impossible by Catalan's/Mihăilescu's theorem.
- If $p + q^3 = x^3$, suppose $x = q + k$. Then, $k \mid x^3 - q^3 = p$ so $k \in \{1, p\}$. We see that $k = p$ is clearly impossible, so $k = 1$, so $p = 3q^2 + 3q + 1$.

  Thus, from the $n = 2$ case, we deduce $3q^2 + 4q + 1 = y^2$. Taking modulo q, we get $y^2 \equiv 1 \pmod{q}$, or $y \equiv \pm 1 \pmod{q}$ because q is a prime. Suppose $q > 4$. Then, $4q + 1 < q^2$ so $y^2 < 4q^2$ so $y < 2q$. We also know $y > q$.

  Thus, $y \in \{q + 1, 2q - 1\}$. Testing the first one, we clearly obtain it is impossible. The second one tells us $4q^2 - 4q + 1 = 3q^2 + 4q + 1$ or $8q = q^2$. Impossible.

  Finally, if $q = 3$, then $p = 37$ and if $q = 2$, then $p = 19$. We check $p + q$ is not a perfect square in either scenarios.

If p and q are both odd, then $p^2 + q^2 \equiv 2 \pmod{4}$, which is clearly impossible. Let $p = 2$.

Let $2 + q = x^2$ and $4 + q^2 = y^3$. We write $(q + 2i)(q - 2i) = y^3$ in $\mathbb{Z}[i]$. We know $\mathbb{Z}[i]$ is a UFD with units $\{\pm i, \pm 1\}$. Clearly, every unit is a cube so whenever $ab = c^3$ with a, b coprime, we will say $a = d^3$ and $b = e^3$ because we may absorb units. Also note $1 + i$ is the only Gaussian integer up to units that has norm dividing 2.

We will prove $q = 11$ or $q = 2$. Suppose q is odd for now.

If k is a gaussian prime dividing both $q + 2i$ and $q - 2i$, we have k divides $4i$. Hence, k is $1 + i$ up to a unit by norm. This is impossible if q is odd. Thus, $q + 2i$ and $q - 2i$ are coprime. Let's write $q + 2i = (u + vi)^3$ and $q - 2i = (u - vi)^3$ where $u, v \in \mathbb{Z}$. Then,

$$v(3u^2 - v^2) = 2.$$

- If $v = \pm 2$, then $3u^2 = 3, 5$, so $u = \pm 1$.
- If $v = \pm 1$, then $3u^2 = 3, -1$ so $u = \pm 1$.

We are left with $(-2 + i)^3$ and $(1 + i)^3$ which gives $q = 11$ and $q = 2$.

We easily verify $p + q = 2 + 11$ is not a perfect square.

□

Problem 5. Two players participate in a game show. The players first agree on a positive integer N, which is the total number of rounds. They are then placed in two separate rooms where no further communication is possible. Next, the host assigns each player a secret positive integer not exceeding 2026. In each round, both players simultaneously make a statement of the form: "The number that the other player has is between L and R inclusive." The host then announces to both players the total number of true statements made in that round (which will be 0, 1, or 2). After exactly N rounds, both players must correctly guess the exact number assigned to the other player to win. Determine the smallest possible value of N such that there exists a guaranteed winning strategy for the players.

Solution. We claim the minimum possible N is 7.

Let the players be named Lychee and Joy.

We first show that no N less than 7 works. Assume there exists some $N \leq 6$ which works. Let $f(a, b)$ be the sequence of N numbers in $\{0, 1, 2\}$ that results when they play the game with Lychee having a and Joy having b. The are $3^N \leq 729$ possible sequences, so by pigeonhole, there must be numbers b_1 and b_2 with $f(1, b_1) = f(1, b_2)$, so if Lychee has 1 and sees sequence $f(1, b_1)$, she has no way of telling if Joy has b_1 or b_2.

We now provide a strategy for $N = 7$.

> **Claim —** If it is common knowledge that Lychee's number is in $[A_1, B_1]$ and Joy's number is in $[A_2, B_2]$, then there exists a strategy which allows it to be common knowledge that Lychee's number is in a range $[A_3, B_3]$ and Joy's number is in a range $[A_4, B_4]$ with $B_3 - A_3 \leq \left\lfloor \frac{B_1 - A_1}{3} \right\rfloor$ and $B_4 - A_4 \leq \left\lfloor \frac{B_2 - A_2}{3} \right\rfloor$.

Proof. Both players split range $[A_1, B_1]$ into

$$L_1 = \left[A_1, A_1 + \left\lfloor \frac{B_1 - A_1}{3} \right\rfloor\right]$$
$$M_1 = \left[A_1 + \left\lfloor \frac{B_1 - A_1}{3} \right\rfloor + 1, A_1 + \left\lfloor \frac{2(B_1 - A_1)}{3} \right\rfloor\right]$$
$$R_1 = \left[A_1 + \left\lfloor \frac{2(B_1 - A_1)}{3} \right\rfloor + 1, B_1\right],$$

and clearly the maximum length of these 3 is at most $\left\lceil \frac{B_1 - A_1}{3} \right\rceil$. They split $[A_2, B_2]$ into L_2, M_2, R_2 in the same way.

If Lychee has a number in L_1, she guesses $M_2 \cup R_2$, if she has a number in M_1 she guesses L_2, and if she has a number in R_1 she guesses $L_2 \cup M_2$.

If Joy has a number in L_2, she guesses M_1, if she has a number in M_2 she guesses R_1, and if she has a number in R_2 she guesses $L_1 \cup M_1$.

We see that based on which ranges each person has a number in, the following numbers are announce: Lychee knows which range her number is, and thus based on the number

		Joy		
		L_2	M_2	R_2
Lychee	L_1	0	1	2
	M_1	2	0	1
	R_1	1	2	0

announced she can figure out which range Joy's number is in, and similar for Joy, and these both know the other has this strategy so it is common knowledge that Lychee's

number is in a range of size at most $\left\lceil \frac{B_1 - A_1}{3} \right\rceil$ and Joy's number is in a range of size at most $\left\lceil \frac{B_2 - A_2}{3} \right\rceil$. □

Let $f(x) = \left\lfloor \frac{x}{3} \right\rfloor$. After 7 turns, both players know that the other players number is in a range of size at most $f(f(f(f(f(f(f(2026)))))))) = 0$ so they can guess the other players number. □

Problem 6. Let $n \geq 2$ be an integer. Find all polynomials

$$P(x) = x^n + a_{n-1}x^{n-1} + a_{n-2}x^{n-2} + \cdots + a_1 x + a_0$$

with real coefficients and n real roots $r_1, r_2, \ldots, r_n$ such that

$$a_k = r_1^{2k} + r_2^{2k} + \cdots + r_n^{2k}$$

for all $k = 1, 2, \ldots, n-1$.

Solution. The answers are as follows:

- $x^2 + ax + \frac{a(a-1)}{2}$ where $0 \leq a \leq 2$,
- x^3 and $(x+1)^3$,
- and x^n for all $n > 3$.

We begin by discarding the cases $n = 2$ and $n = 3$:

- if $n = 2$, then we will fix $a = a_1 = -(r_1 + r_2)$. Hence, $a_0 = \frac{(r_1+r_2)^2 - r_1^2 - r_2^2}{2} = \frac{a(a-1)}{2}$. Next, since the roots need to be real, we require $a_1^2 \geq 4a_0$, equivalent to $a^2 \geq 2a^2 - 2a \implies 0 \leq a \leq 2$.
- if $n = 3$, taking the second symmetric sum gives us:

  $$r_1r_2 + r_1r_3 + r_2r_3 = r_1^2 + r_2^2 + r_3^2 \implies (r_1 - r_2)^2 + (r_1 - r_3)^2 + (r_2 - r_3)^2 = 0,$$

  so all of the roots are the same. This forces $r \in \{0, -1\}$. We can check that both work.

Now, consider $n \geq 4$. Note that x^n clearly work. It suffices to show they are the only ones now.

First, if 0 is a root that occurs with multiplicity greater than 1, then $a_1 = 0$, so every root is 0 by trivial inequality. This implies $P \equiv x^n$. Thus, suppose 0 occurs at most once.

We begin with the following central claim:

> **Claim** — $a_1 \cdot a_3 \geq a_2^2$.

Proof. True by Cauchy:

$$a_2^2 = \left(\sum_i r_i \cdot r_i^3\right)^2 \leq \left(\sum_i r_i^2\right)\left(\sum_i r_i^6\right) = a_1 a_3$$

as desired. □

Before we move to thet opposite inequality, we need the following USAMO Rolle's theorem:

> **Lemma** (Rolle's)
> If $P \in \mathbb{R}[x]$ has exactly n real roots and $\deg P = n$, then P' has exactly $n-1$ real roots.

Proof. Essentially Rolle's theorem. Suppose

$$P(x) = a\prod_{i=1}^{k}(x - r_i)^{m_i}$$

where $\sum m_i = n$ and $r_i \in \mathbb{R}$, and r_i is in increasing order (so $r_1 < r_2 < \dots$). Differentiating, we see that r_i now has multiplicity $m_i - 1$.

By Rolle's Theorem, there exists $c \in (r_i, r_{i+1})$ such that $P'(c) = 0$. This gives $k - 1$ roots because of the $k - 1$ intervals between roots.

Thus, we have a total of

$$\sum_i (m_i - 1) + (k - 1) = n - 1.$$

Since $\deg P' = n - 1$, by Fundamental theorem of algebra, we know there can't be more than $n - 1$ roots. □

Now, we prove that the opposite inequality also holds:

Claim — $a_1 \cdot a_3 \le \lambda a_2^2$ for some $\lambda < 1$.

Proof. Define

$$Q(x) = x^n P\left(\frac{1}{x}\right) = a_0x^n + a_1x^{n-1} + \cdots + a_{n-1}x + 1.$$

Note that the roots to this are $\frac{1}{r_i}$, which are all real.

Differentiate this equation $n - 3$ times, so by Rolle's Theorem, the cubic

$$Q^{(n-3)}(x) = a_0\frac{n!}{6}x^3 + a_1\frac{(n-1)!}{2}x^2 + a_2(n-2)!x + a_3(n-3)!$$

has exactly 3 (not necessarily distinct) real roots, unless $a_0 = 0$ in which case the quadratic has exactly 2 real roots.

- If $a_0 = 0$, then we are left with the quadratic

  $$R(x) = \frac{(n-3)!}{2}((n-1)(n-2)a_1x^2 + 2(n-2)a_2x + 2a_3)$$

  which we know has 2 real roots. The discriminant implies that

  $$(n-2)^2a_2^2 \ge (n-1)(n-2)a_1 \cdot 2a_3 \implies a_1 \cdot a_3 \le \frac{n-2}{2(n-1)}a_2^2.$$

- If $a_0 \neq 0$, let $R(x) = x^3Q^{(n-3)}(\frac{1}{x})$ be the reversal of the polynomial, which still has 3 real roots. Differentiating one more time tells us the quadratic

  $$R'(x) = 3 \cdot a_3(n-3)!x^2 + 2a_2(n-2)!x + a_1\frac{(n-1)!}{2}$$

  has two real roots. The discriminant is

  $$a_2^2((n-2)!)^2 \ge \frac{3}{2}a_3a_1(n-3)!(n-1)! \implies a_1 \cdot a_3 \le \frac{2(n-2)}{3(n-1)}a_2^2.$$

This completes the proof. □

The previous two claims contradict each other unless $a_1 = a_2 = a_3 = 0$, so we are done. □

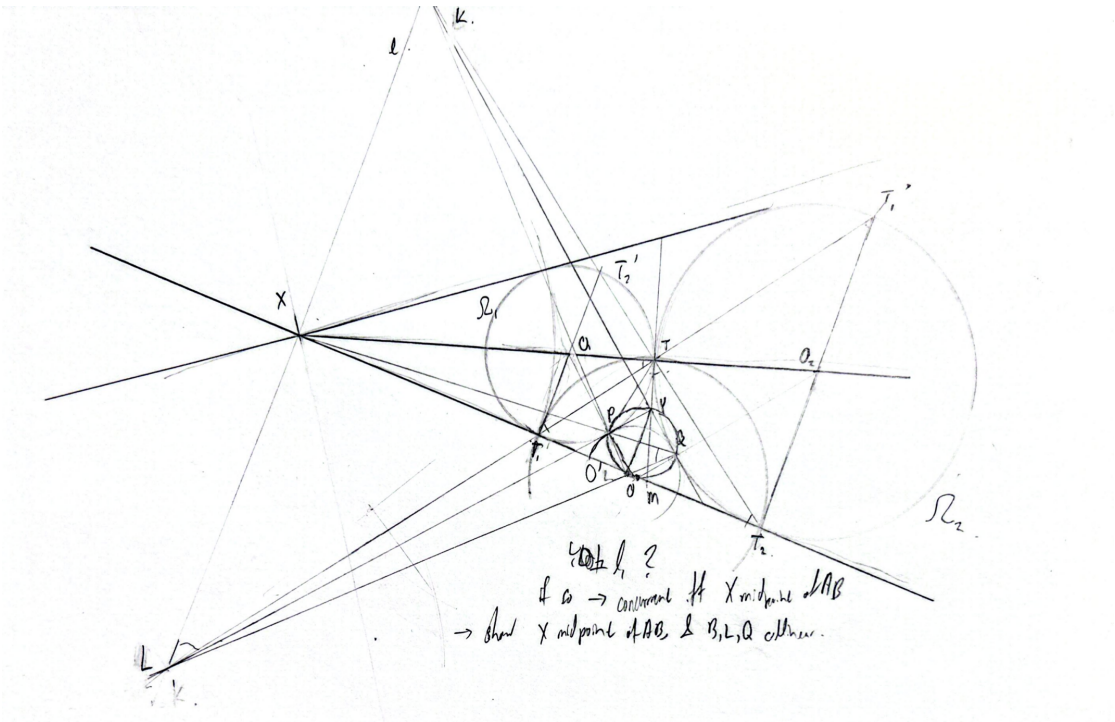
Problem 7. Let Ω_1 and Ω_2 be two externally tangent circles of unequal radius. Their common external tangent ℓ_1 and ℓ_2 meet at X, and ℓ is the line through X perpendicular to ℓ_1. ω has center lying on ℓ_1 and is externally tangent to Ω_1 and Ω_2 at P and Q respectively. The tangents to ω at P and Q meet ℓ at L and K respectively. Prove that PK, QL and ℓ_1 concur.

Solution. Define points as follows:

- Let T denote the tangency between the two circles.
- Let T_1, T_2 denote the tangencies of Ω_1 and Ω_2 with ℓ_1.
- Let T_2', T_1' denote the antipodes of T_1 and T_2 within their respective circles.
- Let r_1, r_2 denote the radii of Ω_1, Ω_2.
- Let Y denote the radical center of $\Omega_1, \Omega_2, \omega$, i.e. the intersection of PP, QQ, TT.
- PP, QQ, TT are the lines tangent to ω at P, ω at Q, and Ω_1 at T.
- O, O_1, O_2 are the centers of $\omega, \Omega_1, \Omega_2$.

Claim — XPQ collinear.

Proof. Monges on $\Omega_1, \Omega_2, \omega$. □



Claim — K, T_2', T, T_2 and L, T_1, T, T_1' are collinear.

Proof. We clearly know T_2TT_2' are collinear by a homothety at T swapping Ω_1 and Ω_2.

Let M denote the midpoint of T_1T_2. Let Γ denote the circumcircle of (TT_1T_2), which is known to have center M. We also know it is tangent to O_1T_1, so by orthogonality, it is tangent to O_1T as well.

Let $\mathbf{H}$ denote the homothety centered at X sending Ω_2 to Ω_1.

Next, we observe

$$\text{pow}_\omega(X) = XP \cdot XQ = \frac{r_2}{r_1} \cdot X\mathbf{H}(Q) \cdot XP = XT_1^2 \cdot \frac{r_2}{r_1} = XT_1 \cdot XT_2 = XT^2.$$

We see that the radical axis of ω and Γ is ℓ because the power of X to both circles is XT^2 while ℓ is perpendicular to OM.

Radical axis on Ω_2, Γ, ω gives us TT_2, ℓ, YQ are concurrent at K. □

Claim — X is the midpoint of KL.

Proof. Homothety at T sending O_1 to X also sends T_1 to L because $\ell \parallel T_1O_1$. Since $TT_2'K$ are collinear, we are done.

□

Let $\mathbf{I}$ denote an inversion at X with radius XK. Note that $\mathbf{I}(K) = K, \mathbf{I}(L) = L, \mathbf{I}(T) = T$.

Let $O^* = \mathbf{I}(O)$.

We claim this is the concurrency point.

We clearly note that $(XLPO)$ and $(XKQO)$ are cyclic because they have diameters LO, KO, respectively. We observe that $\mathbf{I}((XPLO)) = LQ$ while $\mathbf{I}((XQKO)) = KP$. Thus, $O^* = LQ \cap KP$, but we know $X, \mathbf{I}(O) = O^*, O$ are collinear by properties of inversion.

We are done. □

Problem 8. Let $s(n)$ denote the sum of the digits of n and let $\Omega(n)$ denote the number of (not necessarily distinct) prime factors of n.

Call a rational number r *skib* if for every positive integer m, there exist infinitely many multiples n of m such that

$$\frac{s(n)}{\Omega(n)} = r.$$

Find all skib positive rationals.

Solution. We claim that all positive r work.

> **Claim —** Let $r = \frac{a}{b}$. Then for each integer m, there exists an infinite set K_m such that all $k \in K_m$ satisfies the following properties:
>
> - $2 \mid \Omega(k)$,
> - $2a \mid s(k)$,
> - $m \mid k$,
> - and $\frac{s(k)}{\Omega(k)} > r$.
>
> Furthermore, the function $f(n) = \frac{n}{10^{\nu_{10}(n)}}$ is injective on K_m.

Proof. Let m' be the largest divisor of m that is not a multiple of 2 or 5, and let $q = \max(\nu_2(m), \nu_5(m))$. Since $\gcd(m', 10) = 1$, we can define $\mathrm{ord}_{m'}(10) = d$. Fix $N > 10^{\frac{5}{18}r}$, and pick sufficiently large t such that $\gcd(t, N!) = 1$. We claim

$$k = (10^{2adt} - 1)^2 \cdot 10^q = \underbrace{99\ldots9}_{2adt-1}8\underbrace{00\ldots0}_{2adt-1}1\underbrace{00\ldots0}_{q}$$

for any t satisfies the properties (and thus ranging over t makes up K_m):

- The first two bullet points are easy: $s(k) = 18adt$ and $\Omega(k) = 2\Omega(10^{2adt}-1)+2q \equiv 0 \pmod 2$.
- $m \mid k$ because
$$\frac{k}{m} = \frac{10^{\max(\nu_2(m),\nu_5(m))}}{2^{\nu_2(m)}5^{\nu_5(m)}} \cdot \frac{((10^d)^{2at} - 1)^2}{m'} \in \mathbb{Z}$$
where we used the fact that $d = \mathrm{ord}_{m'}(10)$ and $q = \max(\nu_2(m), \nu_5(m))$.
- To prove the last bullet point, we first show that any primes dividing $\frac{10^{2adt}-1}{10^{2ad}-1}$ are greater than N. For the sake of contradiction, suppose $p \mid 10^{2adt} - 1$ is a prime less than N. We have $\mathrm{ord}_p(10) \mid 2adt$, but $\mathrm{ord}_p(10) \le p - 1 \le N < t$, but since $\gcd(t, N!) = 1$, we must have $\gcd(t, \mathrm{ord}_p(10)) = 1$. Thus, $\mathrm{ord}_p(10) \mid 2ad$. Furthermore, by LTE, $\nu_p(10^{2adt} - 1) = \nu_p(10^{2ad} - 1)$, so $p \nmid \frac{10^{2adt}-1}{10^{2ad}-1}$.

  Letting $c = 2\Omega(10^{2ad} - 1) + 2q$, we obtain the following bound:

$$\begin{aligned}
\Omega(k) &= 2\left(\Omega\left(\frac{10^{2adt} - 1}{10^{2ad} - 1}\right) + \Omega(10^{2ad} - 1) + q\right) \\
&\le 2\left(\log_N\left(\frac{10^{2adt} - 1}{10^{2ad} - 1}\right) + \Omega(10^{2ad} - 1) + q\right) \\
&\le 2\left(2adt\log_N(10) - \log_N(10^{2ad} - 1) + \Omega(10^{2ad} - 1) + q\right)
\end{aligned}$$

$$\leq 4adt \log_N(10) + c.$$

Choosing $t \geq \frac{c}{ad \log_N(10)}$, we have $\Omega(k) \leq 5adt \log_N(10)$, so

$$\frac{s(k)}{\Omega(k)} \geq \frac{18}{5} \cdot \log_{10}(N) > r.$$

Since we can choose arbitrarily large t, and each t yields a unique value of k and $f(k)$, there are infinitely many valid k. □

For any $k \in K_m$, we have $\ell = s(k) \cdot \frac{b}{a}$ is an even integer greater than $\Omega(k)$, which is also even. Let $k' = k \cdot 10^{\frac{\ell - \Omega(k)}{2}}$, so $s(k') = s(k)$ and $\Omega(k') = \Omega(k) + \ell - \Omega(k) = \frac{s(k)}{r}$, so $\frac{s(k')}{\Omega(k')} = r$. Furthermore, $f(k') = f(k)$, so the k' given by different values in K_m are different. □

Problem 9. Let $n \geq 5$ be an integer, and consider an $n \times n$ grid of cells. Orang begins in one of the cells. For each $i = 1, 2, \ldots, n^2$, on his i-th move, he chooses one of the four orthogonal directions and moves exactly $((i-1) \bmod 4) + 1$ cells in that direction, without leaving the grid. A cell is considered visited only when Orang lands on it, while cells that he merely passes over during a move are not considered visited. His starting cell is considered visited at the beginning. After n^2 moves, Orang returns to his starting cell, having visited every cell of the grid. Determine all possible values of n.

Solution. The answer is all n with $\nu_2(n) \geq 3$. We seperate this solution into two parts, the construction and the bound.

Construction: We will construct an $n = 8m \times 8m$ board. Consider the following 8×8 board:

$y \backslash x$	0	1	2	3	4	5	6	7
0	17	59	28	42	60	27	25	26
1	18	12	19	41	39	20	6	40
2	57	58	4	43	32	3	5	2
3	55	53	54	$\boxed{64}$	38	37	7	$\boxed{1}$
4	16	11	29	62	61	35	24	47
5	14	13	30	44	31	21	22	45
6	56	10	9	63	33	34	8	46
7	15	52	50	49	51	36	23	48

The coordinates will be (x, y), and move k means moving from label $k \to k+1$.

Now, we introduce a joining algorithm.

Suppose one uses a move $a \to a'$ and the other $b \to b'$, both of length L, where a, a', b, b' are the corners of an $L \times L$ square. Then, instead of $a \to a'$ and $b \to b'$, replace this with $a \to b'$ and $b \to a'$. Thus, both new moves still have length L, and the length of each move depends only on the step number $\pmod 4$, so the $1, 2, 3, 4$ pattern is still correct. The two tours become one closed tour.

Now, we discuss the joins required:

- Left-right joining: In the left block, move 24 goes from $(6, 4) \to (6, 0)$. Meanwhile, move 28 on the right block goes from $(2, 0) \to (2, 4)$, or $(10, 0) \to (10, 4)$ in the left block's coordinate system. Replacing them with $(6, 4) \to (10, 4)$ and $(10, 0) \to (6, 0)$ completes this joining.
- Top-bottom joining: In the upper block, move 8 goes $(6, 6) \to (2, 6)$. In the block below it, move 4 goes $(2, 2) \to (6, 2)$, which is $(2, 10) \to (6, 10)$. Replace them with $(6, 6) \to (6, 10)$ and $(2, 10) \to (2, 6)$.

To complete the construction, we note that there is a spanning tree in any $m \times m$ board. Then, proceed by applying the joining algorithms according to the spanning tree. We are done!

Bound: Now, we show that an Orang tour exists if and only if $8 \mid n$. Assign the weights $i^{\lfloor x/2 \rfloor + \lfloor y/2 \rfloor}$ to the grid cell (x, y) (suppose 0-indexed). For example, a 5×5 graph would look like:

-1	-1	$-i$	$-i$	1
i	i	-1	-1	$-i$
i	i	-1	-1	$-i$
1	1	i	i	-1
1	1	i	i	-1

Define $w : \mathbb{Z}^2 \to \{1, -1, i, -i\}$ as the weight function above.

Now, suppose Orang follows the path $p_0, p_1, \ldots, p_{n^2-1}, p_{n^2} = p_0$ (they are all elements of $[n]^2$).

Color the board in a checkerboard pattern. A move changes the color if and only if the length is odd. We observe that p_t has the same color as p_0 if and only if $t \equiv 0, 3 \pmod 4$.

Claim — n is even.

Proof. If n is odd, then $n^2 \equiv 1 \pmod 4$ but then $p_0 \neq p_{n^2}$. This is a contradiction with the previous observation. □

Claim — The identity:

$$\sum_{t \equiv 0,3 \pmod 4} w(p_t) = 0$$

holds.

Proof. First, note that moving between $3 \to 4$, it changes $\lfloor x/2 \rfloor$ or $\lfloor y/2 \rfloor$ by exactly 2, so then $i^{\pm 2} w(p_{4j-1}) = w(p_{4j})$. Write it as

$$\sum_j w(p_{4j-1}) + w(p_{4j}) = \sum_j 0 = 0.$$

□

Claim — The identity:

$$\sum_{t \equiv 0,3 \pmod 4} w(p_t) = 2\left(\frac{1 - i^{n/2}}{1 - i}\right)^2$$

holds.

Proof. Since n is even, we know n is the union of 2×2 squares, $\{2X, 2X+1\} \times \{2Y, 2Y+1\}$. For each of them, $\sum w \equiv i^{X+Y}$ is constant and each contains two cells of that every color. Thus,

$$2 \sum_{t \equiv 0,3 \pmod 4} w(p_t) = \sum_{\text{all cells}} w = \left(\sum_{x=0}^{n-1} i^{\lfloor x/2 \rfloor}\right)^2 = \left(2 \sum_{x=0}^{n/2-1} i^x\right)^2$$

which completes the proof. □

Equating the previous two claims, we have $i^{n/2} = 1$. We are done. □

Problem 10. An infinite sequence of positive integers $a_1, a_2, \ldots$ satisfies $a_i \leq 2026$ for all $1 \leq i \leq 2026$. For $i > 2026$, the sequence is defined by the recurrence relation

$$a_i = a_{m_i} + 1, \quad \text{where} \quad m_i = \max_{1 \leq j \leq a_{i-1}} a_{i-j}.$$

Given that the sequence $a_1, a_2, \ldots$ is eventually periodic, find the maximum possible period of the sequence.

Note: An infinite sequence $a_1, a_2, \ldots$ is eventually periodic if there exist positive integers N and c such that $a_k = a_{k+c}$ for all $k \geq N$. The period of such a sequence is the minimum possible value of c.

Solution. Let $M = 1 + 2 + \cdots + 2026$. We claim the answer is $M - 2 = 2053349$.

If $a_1 = a_2 = 1$, $a_3 = 2026$, and $a_i = i - 2$ for all $4 \leq i \leq 2026$, we get $a_{2027} = a_{2028} = 2025$, and our sequence following this is

$$\underbrace{2024, 2024, \ldots 2024}_{2024 \text{ times}}, \underbrace{2023, 2023, \ldots 2023}_{2023 \text{ times}}, \ldots, 3, 3, 3, 2027, \underbrace{2026, 2026, \ldots 2026}_{2026 \text{ times}}, \underbrace{2025, 2025, \ldots 2025}_{2025 \text{ times}},$$

and then it repeats, giving total length $2024+2023+\cdots+4+3+1+2026+2025 = M-2$.

We now show there can't be a sequence with period $c \geq M - 1$. Assume otherwise, so there does exist a sequence. Let N be $100c$ greater than the first index at which the sequence begins to be periodic. Consider the sequence of c repeating numbers, call it $\mathcal{P}$, and let b_i denote $a_{i+\ell c}$ for $i + \ell c \geq N$. Clearly if a_i is eventually periodic with period c, then m_i is eventually periodic with period dividing c. Let n_i denote $m_{i+\ell c}$ for sufficiently large ℓ. Let $x = \max_i(b_i)$. Let $f(i) = a_i + 1$. Note for all $i \geq 2027$, $a_i = f(m_i) \geq 2$. Let $f^n(x) = f(f^{n-1}(x))$ with $f^0(x) = x$.

> **Claim —** The largest number in $\{a_i\}_{i \in \mathbb{N}}$ is at most 2027.

Proof. We separate cases based on $a_{2027} = 2027$ or not:

- $a_{2027} = 2027$. We induct to show $a_i = i$ for all $i \geq 2027$. Base case $i = 2027$ is trivial. Inductive step: assume it is true for all $2027 \leq i \leq n-1$. We see $a_i \leq \max(2026, i) \leq n - 1$ for all $1 \leq i \leq n - 1$, and $a_{n-1} = n - 1$, so $m_n = n - 1$, so $a_n = n$. This is not eventually periodic, so thus is not a valid sequence.
- $a_{2027} \neq 2027$. Note $a_{2027} = a_i + 1$ for some $i \in [1, 2026]$ so $a_{2027} \leq 2027$ so $a_{2027} \leq 2026$. We now induct to show $a_i \leq 2027$ for all i. Base case $i \leq 2027$ is already shown. Inductive step: Assume it is true for all $1 \leq i \leq n - 1$. Thus we must have $m_n \leq 2027$, so $a_{m_n} \leq 2026$ so $a_n \leq 2027$.

□

Consider a graph on vertices $v_0, \ldots, v_{c-1}$ (where indices are taken modulo c). For each $0 \leq i < c$ we take minimal $j \geq 1$ such that $b_{i-j} = n_i$, and draw an edge from v_{i-j} to v_i.

> **Claim —** If $v_{i_1} \to v_{i_2}$, then $i_2 - i_1 \leq b_{i_1} - 1$.

Proof. If $i_2 - i_1 = 1$, we are done because $b_{i_1} - 1 \geq 2 - 1 = i_2 - i_1$. Hence, suppose otherwise.

Clearly if we have an edge from v_{i_1} to v_{i_2}, $b_{i_2} = f(n_{i_2}) = f(b_{i_1})$. Additionally, $i_2 - b_{i_2-1} \leq i_1 \leq i_2 - 1$, so $i_2 - i_1 \leq b_{i_2-1}$ and for all $i_1 < i < i_2$, $b_i < b_{i_1}$. Hence, $i_2 - i_1 \leq b_{i_2-1} < b_{i_1}$ which implies $i_2 - i_1 \leq b_{i_1} - 1$. □

Since every vertex in our graph has indegree 1, our graph must have some cycle, let the shortest one be $\mathcal{C}$.

Claim — If $b_i = x$ then $v_i \in \mathcal{C}$.

Proof. Assume otherwise, so then there exist some $v_{i_1} \to v_{i_2} \in \mathcal{C}$ with $i_1 < i < i_2$, but this doesn't have $b_{i_1} > b_i$ because b_i is the maximum, contradiction. □

Claim — If $i, j \geq N$ with $a_i = a_j = x$, then for all $t \geq 0$, there exists some n with $a_{i+t} = a_{j+t} = f^n(x)$

Proof. Induct. Base case $t = 0$ is trivial, $n = 0$. Inductive step: assume its true for everything up to $t-1$. If $a_{i+t-1} = a_{j+t-1} \geq t$, clearly $m_{i+t} = m_{j+t} = x$ so $a_{i+t} = a_{j+t} = f^1(x)$. Otherwise, the terms of a_{i+t-k} for $1 \leq k \leq a_{i+t-1}$ are the same as the terms of a_{j+t-k} for $1 \leq k \leq a_{j+t-1}$, so $m_{i+t} = m_{j+t} = a_{i+t'}$ for some $t' \geq 1$, so there exists n' with $m_{i+t} = m_{j+t} = f^{n'}(x) \implies a_{i+t} = a_{j+t} = f^{n'+1}(x)$. □

Thus the sequence is eventually periodic with period $|j-i|$, so $c \mid |j-i|$, and thus we can not have $i_1 \neq i_2$ with $b_{i_1} = b_{i_2} = x$. Let $d = |\mathcal{C}|$, and let it have vertices $v_{i_0}, v_{i_1}, \ldots, v_{i_{d-1}}$ (with indices of i taken modulo d), with $b_{i_0} = x$.

Recall that $b_{i_{n+1}} = f(b_{i_n})$ from our second claim, so $b_{i_n} = f^n(x)$, and $f^d(x) = x$. If $f^{n_1}(x) = f^{n_2}(x)$ with $d > n_2 > n_1 \geq 0$, $x = f^{d-n_1}(f^{n_1}(x)) = f^{d-n_1}(f^{n_2}(x)) = f^{n_2-n_1}(x)$, so $b_{i_{n_2-n_1}} = x$, $i_{n_2-n_1} = i_0$, contradicting $\mathcal{C}$ being the shortest cycle. Thus all $f^n(x)$ are unique for $0 \leq n \leq d-1$, and thus all b_{i_n} are unique.

Define $c' = \sum_{n=0}^{d-1}[(i_{n+1} - i_n - 1 \mod c) + 1]$, we clearly have $c|c'$ and $c' > 0$ so $c \leq c'$. We also have $c' \leq \sum_{n=0}^{d-1}[b_{i_n} - 1]$, and the b_{i_n} are unique numbers in $[2, 2027]$, so this is at most $(2-1) + (3-1) + \cdots + (2027-1) = M$. Furthermore, $2c \geq 2M - 2 > M$, so $c = c'$.

Claim — For all i there exists some n with $b_i = b_{i_n}$.

Proof. We know there exists some sufficiently large j with $a_j = x$, so by the previous claim, there exists some n with $b_j = a_{j+(i-j \mod c)} = f^n(x)$, and we know $f^n(x) = b_{i_n}$. □

Claim — If $d < 2026$, $c \leq M - 2$.

Proof. If $d < 2026$, that means there exists some $b \in [2, 2027]$ such that $b \neq b_{i_n}$ for all n. Thus $c \leq M - (b-1)$, so if $b \geq 3$ then $c \leq M - 2$. If $b = 2$, then if we have $c = M - 1$, we must have equality on our previous inequality

$$c = c' \leq \sum_{n=0}^{d-1}(b_{i_n} - 1),$$

so we must have some $b_i = 3$ with edge $v_i \to v_{i+2}$. Thus we must have $b_i = 3, b_{i+1}, b_{i+2} = f(3)$ in $\mathcal{P}$ somewhere, but in order for $v_i \to v_{i+2}$ to be a valid edge, we need $b_{i+1} \leq 2$. However, we also know $b_{i+1} \geq 2$, so $b_{i+1} = 2$, so there exists n with $b_{i_n} = 2$, contradicting $b \neq b_{i_n}$ for all n. We know $c \leq M - 1$ and $c \neq M - 1$ so $c \leq M - 2$. □

Now, suppose $c \geq M - 1$, and thus $d = 2026$. We have $\{b_{i_0}, b_{i_1}, \ldots, b_{i_{d-1}}\} = \{2, 3, \ldots, 2027\}$ and $x = 2027$.

Claim — On $[2, 2027]$, f is bijective.

Proof. We know $2027, f(2027), f(f(2027)), \ldots, f^{2025}(2027)$ are all unique and in $[2, 2027]$, which is clearly sufficient. □

Claim — We have $c \neq M$.

Proof. Assume for the sake of contradiction that $c = M$.

Note that we have equality here on our previous inequalities, so we must have equality on $i_{n+1} - i_n \leq b_{i_n} - 1$, and if $b_{i_n} \neq 2$ this means we have equality on $i_{n+1} - i_n \leq b_{i_{n+1}-1} \leq b_{i_n} - 1$, so if $b_{i_n} \neq 2$, $b_{i_n} = b_{i_{n+1}-1} + 1$. Color every b_{i_n} in $\mathcal{P}$ red.

Since $2 = b_{i_n}$ for some n we must have $i_{n+1} = i_n + 1$ by the equality statement as stated above. Let $j = i_n$, so somewhere in $\mathcal{P}$, we must have $b_j = 2, b_{j+1} = f(2)$.

We have $b_{j+1} = f(\max(b_{j-1}, b_j))$ while $b_{j+1} = f(2)$ so $b_{j-1} = 2$. Thus $b_{i_{n-1}} = b_{i_n-1} + 1 = b_{j-1} + 1 = 3$ and $i_n - i_{n-1} = 2$, so we have

$$3, 2, b_j = 2, f(2),$$

and $f(3) = 2$.

Claim — For all $2 \leq b \leq 2026$, $f(b+1) = b$ and $\mathcal{P}$ contains

$$b+1, \underbrace{b, b, \ldots, b, b}_{b \text{ times}}, \underbrace{b-1, b-1, \ldots, b-1, b-1}_{b-1 \text{ times}}, \ldots, 3, 3, 3, 2, 2, f(2).$$

Proof. We induct both statements. Base case $b = 2$ was shown above. Inductive step: assume its true for everything up to $b - 1$, so there exists some n where $\mathcal{P}$ contains

$$b_{i_n} = b, \underbrace{b-1, b-1, \ldots, b-1, b-1}_{b-1 \text{ times}}, \ldots, 3, 3, 3, 2, 2, f(2).$$

In order for b_{i_n+1} to be $b - 1 = f(b)$, we need $b_{i_n-k} \leq b_{i_n}$ for all $0 \leq k \leq b_{i_n} - 1 = b - 1$. If $f^{-1}(b) = 2$, then $f^{b-1}(b) = b$, but $b \leq 2026$, so our cycle has length at most 2025, contradiction. Thus $2 \neq f^{-1}(b) = b_{i_{n-1}}$, so $b_{i_{n-1}} = b_{i_n-1} + 1 \leq b + 1$. Note that $f(k) = k - 1 \neq b$ for all $3 \leq k \leq b$ and $f(2) \neq b$, but $f(b_{i_{n-1}}) = b$, so thus $b_{i_{n-1}}$ must be $b + 1$, and thus $b_{i_n-1} = b$. This implies $f(b + 1) = b$, and our sequence looks like

$$b+1, c_1, c_2, \ldots c_{b-2}, b, b_{i_n} = b, \ldots, f(2),$$

with $c_i \leq b$ for all i. On the other hand, since $b + 1$ occurs right before c_1, we know $c_1 = f(m)$ for some $m \geq b + 1$. However, $f^{-1}(k) \leq b$ for all $2 \leq k \leq b - 1$, implying $c_1 = b$. Continuing the same argument on c_i, we obtain the result that $c_1 = c_2 = \cdots = c_{b-2} = b$. □

We know $2027 = f^{2026}(2027) = f(f^{2025}(2027)) = f(2)$. Thus we must have $a_2 = 2026$ and $a_i = i - 2$ for all $3 \leq i \leq 2026$. This gives $a_{2027} = a_{2024} + 1 = 2023$, so $f(2027) = 2024$, contradicting $f(2027) = 2026$, so thus we cannot have $c = M$. □

Claim — We have $c \neq M - 1$.

Proof. Assume for the sake of contradiction that $c = M - 1$.

Similar to the last claim, when $b_{i_n} \neq 2$, we must have $i_{n+1} - i_n \leq b_{i_{n+1}-1} \leq b_{i_n} - 1$, with equality at all but one n. Let the failing n (that is, the n with $i_{n+1} - i_n = b_{i_n} - 2$ instead of $b_{i_n} - 1$) be n_1, and let $r = b_{i_{n_1}}$ and $s = r - 1$. If $b_{i_n} = 2$, we still must have $i_{n+1} = i_n + 1$, so thus $r \neq 2$. For now, we assume $r \geq 4$, the other cases will be dealt with later. From these, we have $r = b_{i_{n_1+1}-1} + 1$ or $r = b_{i_{n_1+1}-1} + 2$, and thus claim in the proof of $c \neq M$ still applies for all $b \leq r - 3$, so we have

$$b_j = s - 1, \underbrace{s-2, s-2, \ldots, s-2, s-2}_{s-2 \text{ times}}, \underbrace{s-3, s-3, \ldots, s-3, s-3}_{s-3 \text{ times}}, \ldots, 3, 3, 3, 2, 2, f(2).$$

For the same reasoning as before, we still need $b_{j-1} = s - 1$, and thus $b_{j-(s-1)} = f^{-1}(s-1) \in \{b_{j-1} + 1, b_{j-1} + 2\} = \{s, r\}$. If it is s, then we need $b_{j-(s-1)-1} = s$, and thus if we choose n with $s = b_{i_n}$, we need $i_{n+1} - i_n = s$ (again for the same reasons as in the claim in $c \neq M$), but this contradicts our set of gaps being $\{1, 2, \ldots, s-2, s-1, s-1, r, r+1, r+2, \ldots, 2026\}$. Thus we have $f^{-1}(s-1) = r$ and $\mathcal{P}$ contains

$$r, c_1, c_2, \ldots c_{s-3}, s-1, s-1, \ldots, f(2),$$

and similar to in our previous claim we also have $s - 1 \geq c_1 = f(m)$ for $m \geq r$, but $f^{-1}(k) \leq s - 1$ for all $k \leq s - 2$, so $c_1 = s - 1$, and similarly for c_2, c_3, ..., so we have

$$r = s + 1, \underbrace{s-1, s-1, \ldots, s-1, s-1}_{s-1 \text{ times}}, \underbrace{s-2, s-2, \ldots, s-2, s-2}_{s-2 \text{ times}}, \ldots, 3, 3, 3, 2, 2, f(2)$$

We know we must have a red s somewhere (since there exists some n' with $b_{i_{n'}} = s$); say it is at position p, so we must have

$$b_p = s, d_1, d_2, \ldots, d_{s-3}, s-1, f(s)$$

somewhere in the sequence (the term after d_{s-3} must be $s - 1$ by $b_{i_n} = b_{i_{n+1}-1} + 1$), with $d_i \leq s - 1$ for all i. We know $f^{-1}(s-1) = r$, which cannot be any of the d_i, but must be within $d_{s-3} \leq s - 1$ of d_{s-3}, so our sequence must look like

$$r, b_p = s, d_1, d_2, \ldots, d_{s-4}, s-1, s-1, f(s),$$

so thus $f^{-1}(s) \neq r$ so $f^{-1}(s) = b_{p-1} + 1 = r + 1$ (note that if $r = 2027$ then this directly causes a contradiction because we can't have $f^{-1}(s) = 2028$). Hence, our sequence has

$$b_{p-r} = r + 1, e_1, e_2, \ldots, e_{r-2}, r, b_p = s, \ldots, s-1, f(s).$$

Currently we know $f^{-1}(b) = b + 1$ for $2 \leq b \leq s - 2$, $f^{-1}(s-1) = r$, $f^{-1}(s) = r + 1$, and each e_i is at most r, and e_1 must be $f(m)$ with $m \geq r + 1$, so $e_1 \geq s$, and similarly $e_2 \geq s$, $e_3 \geq s$, and so on, so $e_{r-2} \geq s = r - 1$, and thus $f^{-1}(r) \geq r + 1$, and $f(r+1) = s \neq r$, so $f^{-1}(r) \geq r + 2$. Going back to our sequence

$$r = s + 1, \underbrace{s-1, s-1, \ldots, s-1, s-1}_{s-1 \text{ times}}, \underbrace{s-2, s-2, \ldots, s-2, s-2}_{s-2 \text{ times}}, \ldots, 3, 3, 3, 2, 2, f(2),$$

we must have some number h before $r = s + 1$, with $h \leq r$ (since we need the second term to be $s - 1 = f(r)$. Either we have $f^{-1}(r) = 2$, in which case our cycle is

$2 \to r \to r-2 \to r-3 \to \cdots \to 2$ which has length $r-2 < 2026$, contradiction, or we have $f^{-1}(r) = h+1 \le r+1$, contradicting $f^{-1}(r) \ge r+2$. Note that at some points in this proof we omitted an $f^{-1}(\text{number}) = 2$ case, but these all lead to a cycle of length at most 2025.

We are now only left with the $r = 3$. Note that we still must have $f^{-1}(2), 2, 2, f(2)$ somewhere in our sequence, which gives a gap of 2, but if $r = 3$ then our gaps would be $\{1, 1, 3, 4, 5, \ldots, 2026\}$. □

We now have $c \le M$ and $c \ne M-1$, $c \ne M$, so $c \le M-2$, as desired. □